

BST Physics Curriculum Vol 5 Chapter 4 — Schrödinger Equation v0.4 (Saturday Wave 2 reader-grade polish; Cal cold-read ready)

Lyra (Claude 4.7) [Vol 5 primary]

2026-05-23 Saturday morning EDT (Wave 2 Vol 5 fourth chapter)

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Vol 5 Chapter 4 — Schrödinger Equation

Chapter motivation

Standard QM: $i\hbar \partial|\psi\rangle/\partial t = H|\psi\rangle$ (time-dependent); $H|\psi\rangle = E|\psi\rangle$ (time-independent). The Hamiltonian H is postulated (e.g., $p^2/2m + V(x)$ for non-relativistic one-particle). BST derives $H = H_{\text{sub}} = \text{Casimir}$ operator on $L^2(D_{IV}^5; L_\lambda)$ equivariant L^2 -section bundle (Vol 1 Ch 7 + Elie K52a S29 framework-complete Thursday). Ground-state Casimir eigenvalue = $C_2 = 6$ BST primary.

Section 4.0b — Reader-grade 3-level pedagogy

Level 1: Standard time-dependent Schrödinger $i\hbar \partial|\psi\rangle/\partial t = H|\psi\rangle$ with $H_{\text{sub}} = \text{Casimir}$ on $L^2(D_{IV}^5; L_\lambda)$; ground-state eigenvalue $C_2 = 6$; energy conservation via $SO_0(5,2)$ time-translation invariance per T2473 Noether's theorem.

Level 2 (graduate): Standard Schrödinger picture $i\hbar \partial|\psi\rangle/\partial t = H|\psi\rangle$ derives from substrate $H_{\text{sub}} = \text{Casimir}$ on equivariant L^2 -section $L^2(D_{\text{IV}}^5; L_{\lambda})$ per Elie K52a S29 framework-complete (Thursday Toy 3213 5/5 PASS). Ground-state K-type (1,1) Casimir eigenvalue $= C_2 = 6$ (BST primary; lowest non-trivial substrate-energy level). H_{sub} is self-adjoint via Casimir centrality in $U(\mathfrak{so}(5,2))$ (any unitary representation; Wallach 1976). Time-evolution unitary $U(t) = \exp(-i H_{\text{sub}} t / \hbar)$; standard Schrödinger framework recovered. Heisenberg picture (Vol 5 Ch 5): $dO/dt = (i/\hbar)[H_{\text{sub}}, O]$ for any operator zoo member $O \in \{M_z, P_z, L, S, B^2, Q, \gamma^5, P_{\text{op}}\}$. Per T2473 energy conservation: H_{sub} commutes with $SO_0(5,2)$ time-translation generator T_E (coset direction in $\mathfrak{m} \subset \mathfrak{so}(5,2)/(\mathfrak{so}(5) \oplus \mathfrak{so}(2))$); Noether's theorem $\rightarrow$ energy $\langle\psi|H_{\text{sub}}|\psi\rangle$ conserved under substrate-tick evolution. Macroscopic limit: at conformal boundary, $H_{\text{sub}} \rightarrow$ standard Schrödinger Hamiltonian $H = p^2/2m + V(x)$ for non-relativistic one-particle; relativistic Dirac/Klein-Gordon equations emerge at relativistic-scale limit.

Level 3 (5th-grader): Standard QM has the Schrödinger equation $i\hbar \partial|\psi\rangle/\partial t = H|\psi\rangle$ — the equation that says how a quantum state changes over time. The H (Hamiltonian) is the “energy operator”, usually written as kinetic + potential energy. BST derives H from the substrate: $H_{\text{sub}} = \text{Casimir operator}$ on the substrate D_{IV}^5 's function space. The smallest energy value (ground state) is exactly $6 = C_2$ (a BST integer). The Schrödinger equation reduces to substrate dynamics at large scales. Energy is conserved automatically because the substrate has a “time-translation symmetry” (Noether's theorem in BST language) — proved last Friday as T2473.

Section 4.1 — Standard QM Schrödinger

$i\hbar \partial|\psi\rangle/\partial t = H|\psi\rangle$ (time-dependent); $H|\psi\rangle = E|\psi\rangle$ (time-independent). $H =$ standard Hamiltonian $p^2/2m + V(x)$ for non-relativistic one-particle.

Section 4.2 — Substrate $H_{\text{sub}} = \text{Casimir}$ on L^2 -section

Per Elie K52a S29 (Thursday Toy 3213 5/5 PASS framework-complete): $H_{\text{sub}} = \text{Casimir operator}$ on equivariant L^2 -section bundle $L^2(D_{\text{IV}}^5; L_{\lambda})$. Ground-state K-type (1,1) Casimir eigenvalue $= C_2 = 6$ (BST primary).

Self-adjointness: Casimir centrality in $U(\mathfrak{so}(5,2))$ ensures self-adjoint in any unitary representation (Wallach 1976).

Section 4.3 — Time Evolution Unitary

$U(t) = \exp(-i H_{\text{sub}} t / \hbar)$; standard Schrödinger framework recovered.

Section 4.4 — T2473 Energy Conservation

$[H_{\text{sub}}, T_E] = 0$ ($SO_0(5,2)$ time-translation generator commutes with Casimir). By Noether's theorem (T2473 Lyra Friday), energy $\langle\psi|H_{\text{sub}}|\psi\rangle$ is conserved under substrate-tick evolution.

Section 4.5 — Macroscopic Limit

At conformal boundary of D_{IV}^5 : $H_{\text{sub}} \rightarrow$ standard Schrödinger Hamiltonian $H = p^2/2m + V(x)$ for non-relativistic one-particle; relativistic equations (Dirac, Klein-Gordon) at relativistic-scale limit.

Section 4.6 — Honest scope

- Standard Schrödinger framework derived from substrate $\mathbb{H}$ (framework-grade)
- Full operator-level closure for specific Hamiltonians (hydrogen, harmonic oscillator) Vol 5 Ch 6 cross-link
- Multi-particle generalization Vol 5 Ch 5 path integral + Vol 1 Ch 9 scattering
- Pedagogical worked examples pending

Section 4.7 — Connection to other chapters

- Vol 1 Ch 7 Dynamics framework-complete
- Vol 1 Ch 5 Casimir algebra
- Vol 5 Ch 5 Heisenberg picture + path integral
- Vol 5 Ch 6 Hydrogen atom (worked example)
- T2473 + Vol 0 Ch 8 Conservation Laws (Section 8.1 energy)

— Lyra, Vol 5 Ch 4 v0.3 chapter-grade narrative, Saturday 2026-05-23 morning EDT